6. Function

Hengfeng Wei (魏恒峰)
hfwei@hnu.edu.cn



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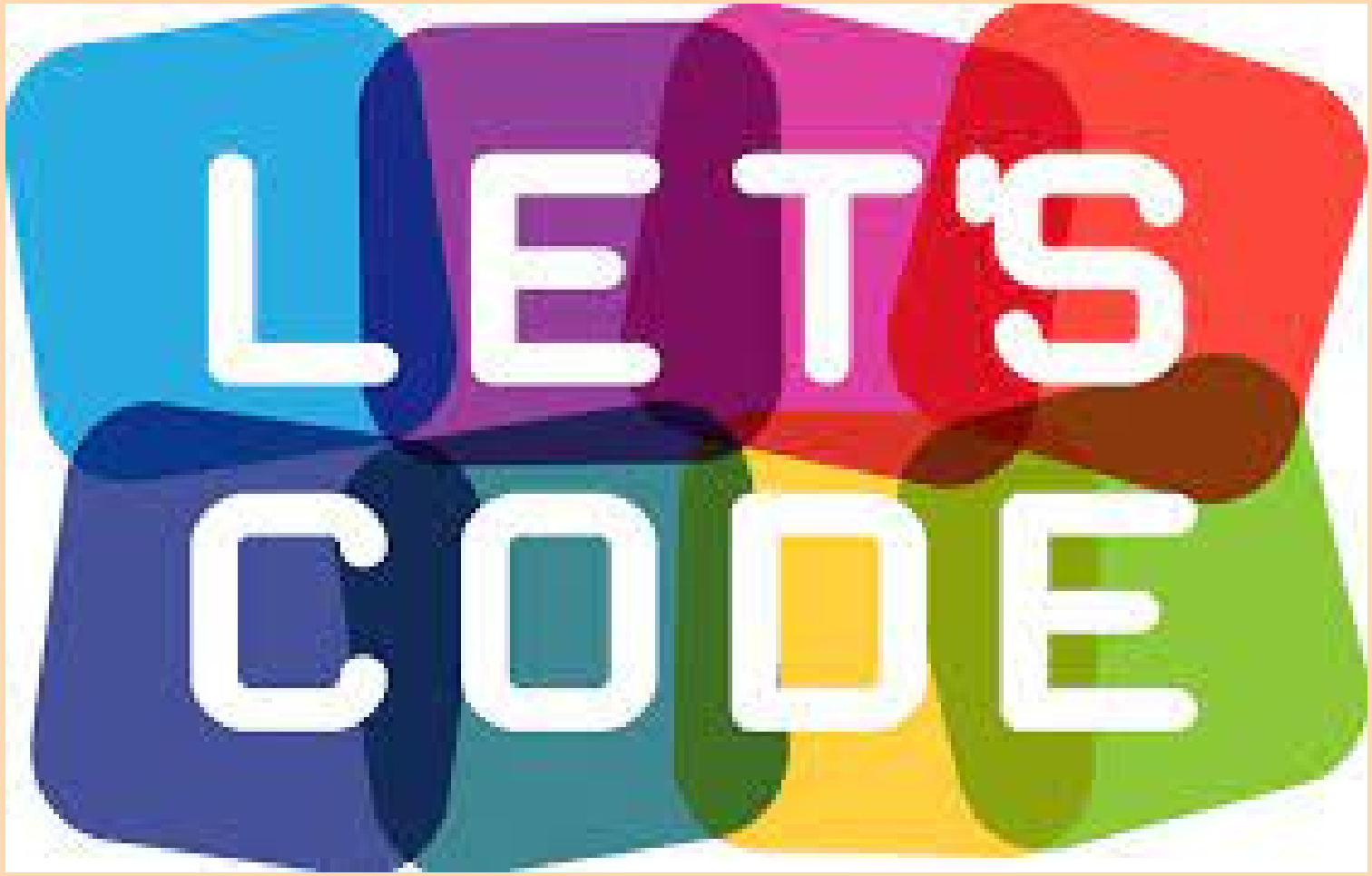
Overview

Functions

Function Definition

Function Declaration

Pass by Value/Reference/Const Reference


LET'S
CODE



Re-writing Programs using Functions

Using std::span Well

When to use it, and what to watch out for

C++20 STANDARD LIBRARY

Use it when...

- A function needs a contiguous sequence
- The function should accept C arrays, `std::array`, or `std::vector`
- You need read-only access: `std::span<const T>`
- You need in-place updates: `std::span<T>`

Do not expect...

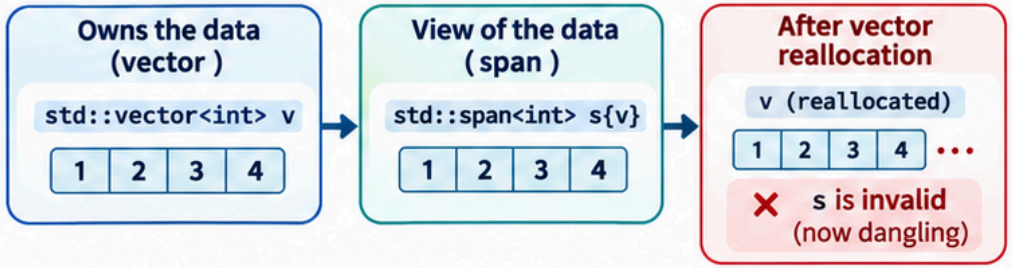
- Ownership of the data
- Automatic lifetime extension
- Resizable storage (no `push_back`, no `resize`)
- Universal compatibility with every container

Container compatibility

- ✓ `C array`
- ✓ `std::array`
- ✓ `std::vector`
- ✗ `std::list`
- ✗ `std::deque`
- ✗ `std::set`
- ✗ `std::vector<bool>`

span requires contiguous elements

Key safety rule



- The underlying storage must outlive the span
- Vector reallocation invalidates existing spans
- Bounds still matter: avoid out-of-range indexing

Rule of thumb: prefer `std::span` for function parameters that need a view of contiguous data.


THANK YOU!